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REPORT

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Mannose receptor contributes to preferential oligomannosidic glycoform clearance in a murine pharmacokinetics model

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ABSTRACT

The pharmacokinetic profile of therapeutic antibodies, which affects the dosing frequency, efficacy, and cost of the products, is influenced by several molecular features, including N- glycosylation. While the accelerated clearance of antibodies carrying non-complex type glycans is well established, the mechanism behind this phenomenon is poorly understood. Though there have been indirect indications before, we provide the first direct evidence for the involvement of the mannose receptor in the clearance of therapeutic antibodies. Wild-type (WT) mice showed accelerated clearance of an oligomannose enriched antibody, and this effect was reduced in mannose receptor (MRC1) knockout (KO) mice. Subsequent glycoform-resolved analysis by mass spectrometry showed in WT an accelerated clearance of oligomannosidic compared to complex glycoforms. This accelerated clearance was also observed in MRC1 KO mice, but to a lesser extent. Interestingly, the glycoform distribution of total circulating mouse antibodies, which do not contain oligomannosidic glycoforms, was not affected by MRC1 KO. Hence, we show that the mannose receptor is involved in, but not exclusively responsible for, preferential clearance of oligomannosidic glycoforms of antibodies. This points toward a role for additional, glycan-targeting receptors for therapeutic antibody clearance. In addition, we observed a preferential clearance of complex afucosylated glycoforms, unique to the mouse model, which was MRC1-independent.

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Introduction

Protein glycosylation is a common and complex post-translational modification with prominent impact on turnover, stability, and receptor–ligand interactions of glycoproteins. Therefore, it is a critical quality attribute (CQA) for many therapeutic glycoproteins.[1,2] Glycosylation is a key factor in the efficacy of therapeutic monoclonal antibodies (mAbs) and antibody-derived formats, whose >200 billion USD market share makes them the most important class of biologicals. Amongst several other factors, glycosylation has been shown to impact antibody clearance and plasma half-life.[3,4] Specifically, clearance can be more than doubled for glycoforms presenting outer arm mannoses (hybrid type and oligomannose glycans) on their conserved crystallizable fragment (Fc) glycosylation site, when compared to complex diantennary glycoforms.[4–6] While the increased clearance of mAb oligomannosidic glycoforms is overall well established, the mechanism behind it is far less well explored.

Glycan moieties are crucial when it comes to glycoprotein homeostasis and clearance from the circulation. Modulating glycan patterns of glycoproteins has been shown to affect their concentrations, as well as rate of clearance from the circulation.[1] The affinity for glycan-binding proteins with strong motif preferences governs the selective recognition, internalization, and clearance of glycoproteins from the circulation. C-type lectins (CTLs), which contain a Ca^{2+}-dependent

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carbohydrate recognition domain, are the largest glycan-binding protein family that interacts with circulating glycoproteins.[2] The asialoglycoprotein receptor and the mannose receptor C-type 1 (MRC1) are the two CTLs most often suggested to mechanistically contribute to glycoform-selective clearance of circulating glycoproteins.

MRC1, also called the mannose receptor or CD206, is an endocytic receptor expressed by subsets of macrophages, dendritic cells, and nonvascular endothelial cells.[7,8] It is important in pattern recognition, modulation of antigen presentation, cellular activation, and trafficking induced by canonical pattern recognition receptors. In addition, MRC1 has roles in the homeostasis of endogenous molecules, such as collagen. MRC1's CTL domain is assumed to play the largest role in its involvement in the pharmacokinetics (PK) of therapeutic proteins. Classically, the CTL is reported to have affinity for outer arm mannoses, terminal *N*-acetylglucosamine (GlcNAc), and fucose-containing Lewis-type structures. Recently, MRC1 glycosylation has been demonstrated to impact its lectin activity and substrate specificity.[9] Furthermore, for mAb Fc-glycans, interaction has been suggested to be limited to mannoses, and enhanced if both heavy chains present oligomannosidic glycans.[10] Furthermore, MRC1 contains a fibronectin type II domain for collagen binding and a cysteine-rich domain binding 3- or 4-sulfated galactose and N-acetylgalactosamine. Genetic variants of MRC1 have been associated with several diseases, including asthma and leprosy,[11,12] suggesting that there might be functionally different variants across the population, which could also influence PK behavior of therapeutic proteins.

Preferential clearance of outer arm mannose-containing glycoforms of mAbs and other therapeutic proteins is well established. Yet, mechanistic studies into this phenomenon are scarce. One early study found that mannan could reduce the clearance of oligomannosylated mAbs, suggesting the involvement of oligomannose-targeting pattern recognition receptors.[13] Furthermore, the affinity of oligomannosidic mAb glycoforms to truncated and full-length versions of the extracellular domain of MRC1 has been established by biophysical methods.[10,14] However, given the importance of exposed mannoses in distinguishing self from foreign glycoproteins, a large variety of immune receptors target mannose motifs,[15] and no in vivo evidence is available linking the increased clearance of oligomannosylated mAbs to a specific receptor.

Here, we aimed to gather direct evidence for the involvement of MRC1 in the preferential clearance of outer arm mannose containing antibody glycoforms (see Table S1 for nomenclature). We monitored the clearance of two mAbs in a glycoform-resolved manner from wild-type (WT) and MRC1 knockout (KO) mice (Figure 1). Both antibodies had the same amino acid sequence with a target of which a homologue was present in the model system. However, the antibodies do not bind the homologue. Both antibodies were expressed in Chinese hamster ovary (CHO) cells, but one was enriched in oligomannosidic glycans. Changes in the glycoprofiles of the mAbs were monitored using the GLYcoLISA methods as previously demonstrated.[5,16] In addition, we monitored the natural IgG antibodies of the model system, allowing us to assess potential systemic effects of receptor KO on antibody processing.

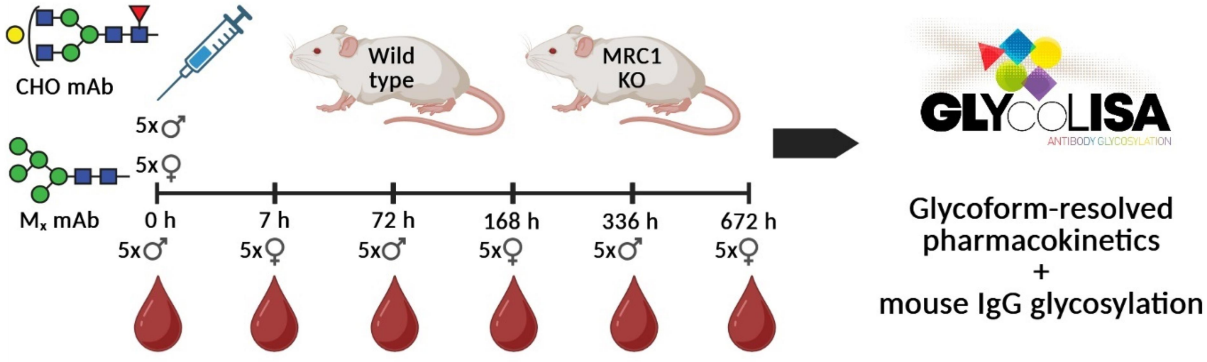


Figure 1. Setup of the single dose pharmacokinetics study of the two mAbs. For each of the four groups, 5 male and 5 female mice were injected at 0 h, and then bled at alternating time points. Figure created in BioRender.

Results

MRC1 deficiency reduces preferential oligomannose clearance, but does not abrogate it

We aimed at quantifying the role of MRC1 in the clearance of mAb glycoforms with outer arm mannose residues by comparing the clearance of mAbs in WT and MRC1 KO mice. For WT mice, we observed the expected preferential clearance of oligomannosidic glycoforms leading to a reduction in their relative abundance over time compared to complex type glycoforms (Figure 2 and Table S2). We observed this for both a mAb with the common glycosylation profile associated with CHO cell expression (CHO mAb; Figure S1 and Figure 2(A)), as well as for a mAb glycoengineered to be enriched in oligomannose glycans (M_X mAb; Figure S1 and Figure 2(B)). The MRC1 KO mice also showed a preferential clearance of oligomannosidic glycoforms of both mAbs. However, the change in glycosylation profile in favor of the complex type glycans was less pronounced. For example, in WT mice injected with the CHO mAb, oligomannose content was 29% (0.95%±0.14% versus 3.33% ±0.06%) of the initial relative abundance after 336 h while in the MRC1 KO mice this only changed to 57% (1.84%±0.10% versus 3.24%±0.26%). The study arm with M_x mAb confirmed this observation with 24% versus 49%. Thus, MRC1 was responsible for a part of the preferential clearance of oligomannosidic glycoforms in WT mice, but removal of MRC1 did not abrogate the effect entirely in KO mice.

Biotransformation by circulating mannosidases can be saturated

Man5 was the main oligomannosidic glycoform observed in the CHO mAb with minor contributions of Man6, Man7 and Man8 (Figure S1, Table S3 and Table S4). As expected, through the action of circulating mannosidases, the higher oligomannosidic glycoforms rapidly disappeared and where not detected after 7 h in the CHO mAb (Figure S2). This was not impacted by the KO of MRC1. In the M_X mAb, Man5 to Man9 were present with Man8 and Man9 as major glycoforms, and Man5 to Man7 as minor glycoforms (Figure S1, Table S3 and Table S4). Man7 to Man9 were still detectable after 72 h, and Man6 remained present for the entire monitoring period of 672 h or 28 days (Figure 1(E) and S2). In the M_x mAb, Man 5 and Man 6 were likely generated in vivo from Man 7 to Man 9, and accumulated relative to other glycoforms as the biotransformation is faster than the preferential clearance. In the KO mice, this was more pronounced due to the reduced clearance. Thus, though MRC1 KO did not impact biotransformation by mannosidases, glycosylation profiles differed between WT and KO mice from 72 h on. Of note, the ratio of Man5/Man6 was not impacted.

MRC1 deficiency does not affect glycosylation of natural mouse antibodies

We expected the changes in relative abundance of oligomannosidics over time and differences between WT and MRC1 KO mice to be governed by clearance effects. To further exclude an impact of potential in vivo glycan remodeling on our findings, we analyzed the natural mouse antibodies (mIgG) in WT and MRC1 KO mice (Tables S5 to S8). We did not observe any differences in mIgG glycosylation between WT and KO mice, as exemplified by the most abundant subclass mIgG2ac in Figure S3. In addition, mIgG glycosylation was not impacted by mAb glycosylation, meaning it did not differ between animals administered with CHO mAb or M_X mAb (data not shown).

Preferential clearance of monoantennary and afucosylated, complex glycoforms is not affected by MRC1 deficiency

We also monitored several glycosylation features of complex type glycoforms in the mAbs and observed a number of interesting behaviors. The monoantennary glycoforms (Figure S4B+C) showed the expected preferential clearance which, however, was not impacted by MRC1 KO. This extends the observation that monoantennary mAb glycoforms do not directly bind MRC1, but provides indication against the suggested oligomannose pairing preference-mediated mechanism.[10] Not reported previously, we observed a preferential clearance of complex type, afucosylated mAb glycoforms that was equally MRC1 independent (Figure 3). As a negative control, complex type galactosylation of the mAbs was constant over time and did not differ between WT and MRC1 KO experiments (Figure S4 C and D).

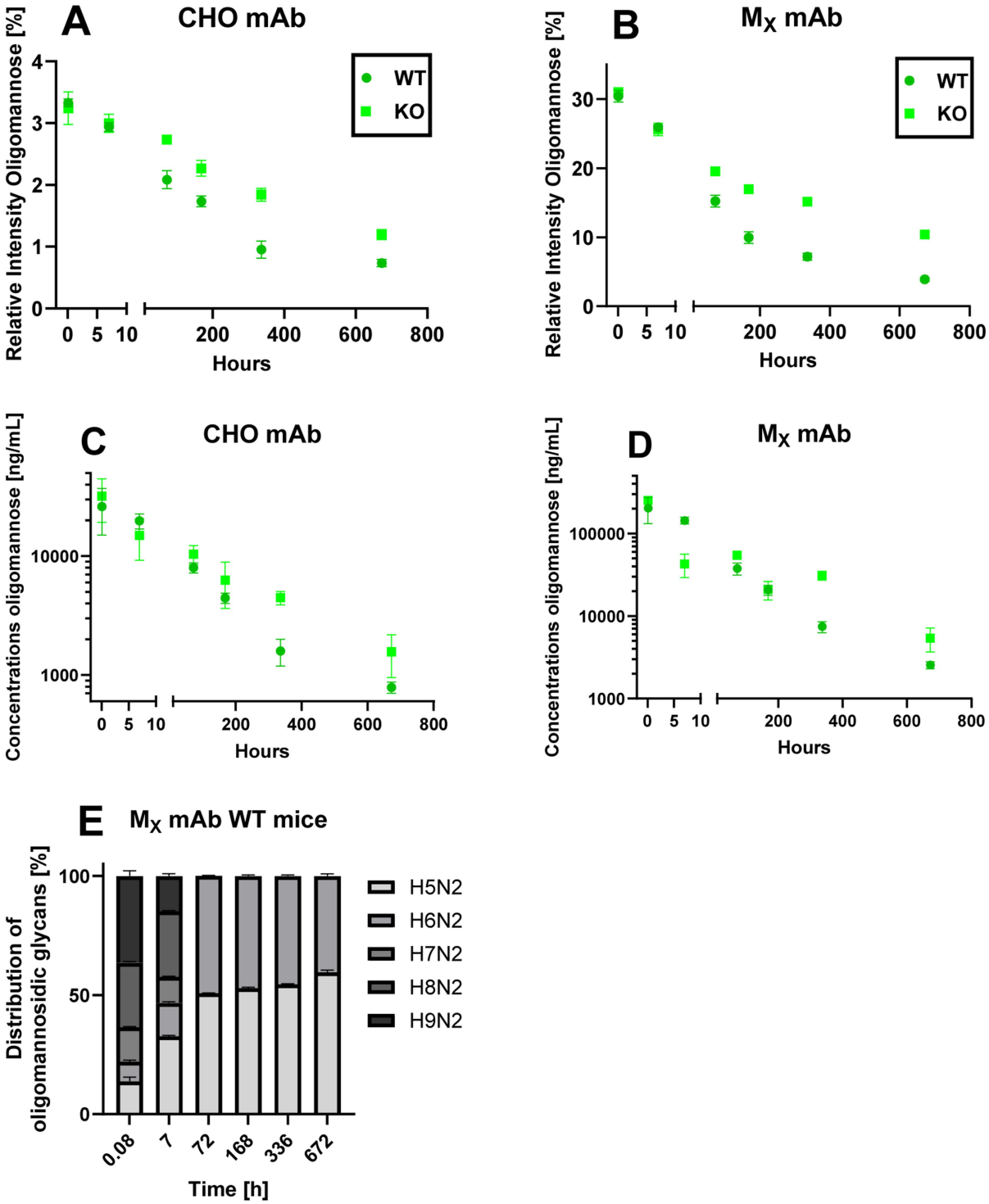


Figure 2. Change in relative and absolute abundance of oligomannosidic glycoforms over the course of the clearance of two mAbs in wild type (wt) and mannose receptor C-type 1 knockout (KO) mice. A+C) mAb with a standard CHO cell type glycosylation (CHO mAb; Figure S1) and B+D) glycoengineered mAb with increased oligomannose glycans (M_x mAb; Figure S1). e) changes in the compositions of oligomannosidic glycans due to biotransformation (see also Figure S2). Mean and standard deviation ($n = 5$) are depicted with sd often smaller than point size.

Glycoform-resolved pk parameters support findings from longitudinal relative intensity changes

Table 1 shows the clearance values of individual glycoforms. The color scale highlights whether the values are increased or reduced relative to the reference glycoform G0F. Note that some animals were removed from this analysis due to unexpected PK behavior (see Experimental procedures). Oligomannosidic glycoforms were cleared significantly faster than G0F, also in the KO mice. However, oligomannose clearance was higher in WT mice than in KO mice (CHO mAb: 0.60 ± 0.14 versus 0.42 ± 0.12, $p = 0.012$; M_X mAb: 0.87 ± 0.13 versus 0.57 ± 0.06, $p = 0.0005$). Accordingly, half-life (Table S9) was decreased, more in WT mice than in KO mice (WT versus KO: CHO $p = 0.048$, M_X $p = 0.011$). Of note, due to the enzymatic biotransformation of higher oligomannosidic glycoforms to Man 5 and Man 6, Man 5 effects appeared lower than the sum of the oligomannosidic species, to the point where Man5 misleadingly showed a decreased clearance (increased AUC in Table S10) values in the M_x mAb. Interestingly, half-life calculations seemed less affected. Also, monoantennary (G0-N, G0F-N and G1F-N) and afucosylated (G0-N, G0, and G1) glycoforms showed increased clearance (decreased half-life and AUC) of clearly additive nature, as the combined G0-N showed highest clearance in these groups. KO did not impact the monoantennary or

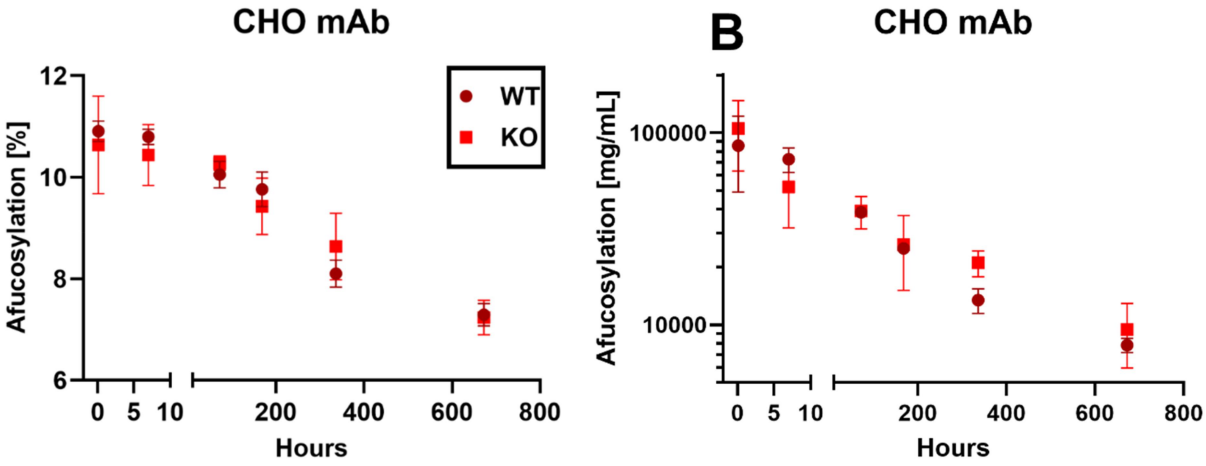


Figure 3. Time-dependent changes in complex type afucosylation of the CHO mAb. Complex type, afucosylated mAb glycoforms are also cleared faster than their fucosylated counterparts. a) relative abundance and B) concentration changes. Mean and standard deviation are depicted which sd often smaller than point size.

Table 1. Clearance/Dose [mL/h/kg].

Glycoforms	CHO mAb		M_X mAb	
	WT mice	KO mice	WT mice	KO mice
G0F	0.26±0.05 (9)	0.22±0.08 (8)	0.26±0.06 (9)	0.20±0.02 (3)
G0-N	0.50±0.11 (9)***	0.43±0.13 (8)***	0.58±0.13 (9)***	0.49±0.06 (3)***
G0F-N	0.27±0.06 (9)	0.26±0.09 (8)*		
G1F-N	0.28±0.06 (9)	0.24±0.08 (8)		
G0	0.35±0.07 (9)***	0.30±0.10 (8)***		
G1	0.29±0.06 (9)*	0.26±0.09 (8)**		
G1F	0.25±0.05 (9)	0.22±0.08 (8)	0.23±0.06 (9)	0.18±0.02 (3)
G2F	0.26±0.05 (9)	0.23±0.08 (8)	0.24±0.06 (9)	0.18±0.02 (3)
Man 5	0.42±0.10 (9)***	0.29±0.08 (8)***	0.19±0.04 (9)**	0.08±0.02 (3)***
Man6			0.16±0.01 (5)	
Oligomannose	0.60±0.14 (9)***	0.42±0.12 (8)***	0.87±0.13 (9)***	0.57±0.06 (7)***
Total mAb	0.28±0.05 (10) ♂0.33±0.04 (5) ♀0.24±0.02 (5)	0.22±0.03 (7) ♂0.24±0.01 (4) ♀0.19±0.02 (3)	0.39±0.09 (9) ♂0.49±0.05 (4) ♀0.32±0.01 (5)	♂0.31±0.03 (5)

Fold change versus G0F: 0.4 1.0 3.3 ; mean ± standard deviation (number of animals; see Methods); *p < .05; **p < .005; ***p < .0001; [1]not tested.

afucosylation effects. If anything, the monoantennary effect was more visible against the reduced oligomannose effect in the KO mice. Finally, galactosylation did not affect PK parameters, as these were not different for G1F and G2F compared to G0F. For completeness, we also calculated c_{max} (Table S11).

Differences in clearance were also visible on the total mAb level (Table 1 last row). As expected, clearance was reduced in KO compared to WT mice for both mAbs (CHO ♂ $p = 0.006$ and ♀ $p = 0.04$; M_X $p = 0.001$). Orthogonally, clearance of M_X mAb was higher than that of CHO mAb (WT $p = 0.0005$ and KO $p = 0.02$). Effects on other PK parameters could only be observed incidentally.

Discussion

Preferential clearance of oligomannosidic and hybrid type glycoforms is the strongest and most well-established impact of glycosylation on mAb PK. It was shown in a mouse model that this effect can be inhibited by mannan,[13] but no direct evidence had been presented for the involvement of a specific clearance receptor. A previous study showed an impact of MRC1 knockout on glycoprotein clearance in a mouse model, but using bovine serum albumin modified with a single mannose residue.[17] This highly artificial system for both glycan and protein context is unlikely to comprehensively reflect the binding epitope and accessibility, especially for the relatively shielded mAb glycosylation site.[18] Our PK study in MRC1 KO mice confirmed previous expectations, but also revealed unexpected insights regarding the involvement of MRC1 in the glycoform-specific clearance of mAbs. We provided strong evidence of the involvement of MRC1 in the preferential clearance of oligomannosidic antibody glycoforms. Furthermore, we also demonstrated that the involvement of MRC1 is strong enough to impact the overall PK behavior of an mAb. Most importantly, we showed that MRC1-mediated clearance is not the only mechanism by which oligomannosidic glycoforms are preferentially cleared, although it is a strong contributor. Interestingly, oligomannosidic glycoforms of natural mouse plasma proteins are not affected by MRC1 deletion as we have previously demonstrated.[15] Therefore, mAbs probably present a distinct, yet highly relevant, scenario when it comes to the role of MRC1. When trying to extrapolate the results of this study, potential differences between mouse and human MRC1 in terms of structure, glycan-binding preferences, and cell-specific expression, next to differences in immune cell composition must be considered.

In this light, the absence of an effect on glycoforms with terminal GlcNAc is remarkable, observed as stable terminal galactosylation. Preferential clearance of terminal GlcNAc-containing glycoforms has been reported for Fc fusion proteins containing several accessible glycans,[3] but, for mAbs with only shielded Fc glycans, only very minor effects have been reported.[4] Thus, our study confirms previous findings.

The M_X mAb, as an antibody with mainly larger oligomannosidic glycoforms, allowed the recapitulation of earlier findings regarding trimming of oligomannosidic glycans,[19,20] as well as the observation of novel results. We conclude that MRC1 does not seem to have a preference for a lower or higher number of outer arm mannoses, as the oligomannosidic profile changes were highly comparable between WT and KO mice. Through a combination of high antibody concentrations and high oligomannose content of the M_X mAb, we made observations that indicate the enzymatic system responsible for trimming larger oligomannosidic glycoforms to Man5,[6] has a limited capacity that can be saturated with therapeutic proteins. This should have implications for the physiological role of this system and possibly its interaction with oligomannose-shielded pathogens, such as HIV.[21] While previously published data show the same phenomenon,[19,20] previous discussions did not highlight this point. Potentially, the phenomenon was not as obvious, due to lower oligomannose content and sample numbers.

Interestingly, we observed preferential clearance of complex type, afucosylated mAb glycoforms that was independent of MRC1. This effect was additive to the previously described effects of monoantennary glycans, as demonstrated by the strong clearance of G0-N.[4] This is in contrast to porcine model systems that do not show such a preference upon intravenous (IV) injection,[4] and previous rodent studies that showed only small effects.[5,6] Due to its strong preferential binding to afucosylated IgG glycoforms, the Fc gamma receptor (FcγR) III, or in this case its mouse homologue FcγRIV,[22] is an obvious agent for mediating such a preferential clearance. At high mAb concentrations, however, FcγR-mediated clearance is not expected to impact PK behavior significantly, as demonstrated in cynomolgus monkeys.[23] Natural rodent IgG is (almost) completely fucosylated, which may have implications for the activation threshold of mFcγRIV, potentially

making rodents more susceptible to the effects of afucosylation. Though it was shown that general binding strength and preference for afucosylated glycoforms is preserved in the interaction of human IgG with mFcγRIV compared to hFcγRIII,[22,24] immune cell populations and expression levels of FcγR differ largely between species. For example, neutrophils are much less abundant in circulation of mice compared to humans and there is no homologue for hFcγRIIIb, the most prevalent human FcγR. The high stability of the complex between afucosylated IgG and hFcγRIIIa has been demonstrated *ex vivo*.[25] This clearance phenomenon is target-independent, since the mAbs do not bind the mouse homologue of the human mAb target. These aspects could provide an explanation for the quantitative differences in findings compared to earlier studies. Whether binding to FcγR significantly contributed to clearance in our study needs to be further investigated. However, it could be a contributing factor to the residual, MRC1-independent, preferential clearance, as oligomannosidic glycoforms also have a strongly enhanced affinity for FcγRIII.

Glycoengineering for reduced fucose content is a popular, efficient, and successful means to generate biobetter mAbs with enhanced FcγRIII-mediated effector functions.[26] Use of kifunensine and swainsonine supplementation, creating oligomannose and hybrid-type glycosylation, respectively, has been proposed for increasing oligomannose content in early studies, but this has been abandoned due to the negative impact on PK behavior.[6] Supplementation with decoy substrates, FUT-8 knock-out, or the GlycArt technology are popular approaches because they generate largely complex, afucosylated glycoforms. Our findings regarding the preferential clearance of afucosylated species specifically in a mouse model suggest a risk that mouse models may uniquely show an unfavorable PK behavior for glycoengineered, afucosylated antibodies.

As we have speculated previously, there are a number of alternative C-type lectins with oligomannose specificity that decorate macrophages and other immune cells.[15] These include mannose-binding lectin (MBL), macrophage C-type lectin (MCL), macrophage inducible C-type lectin (MINCLE) and dendritic cell-specific intercellular adhesion molecule-3-grabbing nonintegrin (DC-SIGN). CTLs on immune cells are prime candidates to explain the residual preference for oligomannosidic glycoform clearance in the absence of MRC1, as they would likely be able to access very similar mechanisms and pathways for clearance. In the future, glycoform-resolved biophysical and cellular-binding studies will have to further clarify the mechanism for oligomannose mAb glycoform clearance by MRC1 and alternative mAb clearance receptors.[10,27]

Several genetic variants of human MRC1 are known and have been associated with diseases, suggesting strong interindividual differences in MRC1 function and/or expression. Due to the strengthened link of MRC1 with mAb PK properties, it would be interesting to study the impact of MRC1 genetic variants on mAb PK in the future. Should genotypes impact clearance, ultimately, genotyping of MRC1 and other CTLs could contribute to determining optimal, personalized dosing of mAbs.

Materials and methods

Glycoengineering

CHO mAb and M_x mAb had an IgG1 heavy chain, shared the same amino acid sequence, and were produced in CHO cells. A batch of mAb was glycoengineered by adding 0.1 µg/mL kifunensine to the basal medium of the fermentation process as previously described.[4] However, as opposed to previous work, the mAb was not further processed with mannosidase A, thus retaining larger oligomannosidic structures than previously reported. M_x mAb was then obtained by mixing one part of this preparation with two parts of the CHO mAb.

Single-dose pharmacokinetic study in mice

Animal experiments were approved by the Ethic Committee of Toulouse Biological research Federation (C2EA-01) and by the Ministry of Higher Education and Research (APAFIS#30917). C57BL/6J WT and MRC1 KO (B6.129P2-Mrc1tm1Mnz/J) mice,[17] backcrossed with C57BL/6J mice for six generations, were bred in a specific pathogen-free zone within the animal facility of the Institute of Pharmacology and Structural Biology. MRC1 KO was confirmed by genotyping.

The study consisted of four arms, two with WT and two with MRC1 KO mice, who received either CHO mAb or M_x mAb. WT and MRC-1 KO mice aged between 8 and 12 weeks were anesthetized (induction box: 4%, induction mask: 2%; Isovet OSALIA), and then 120 µL of either mAb formulation were administered by IV injection into the tail vein. mAbs were formulated in a solution of 100 mM sodium acetate (pH 5.0) at 7.0 mg/mL for CHO mAb and 7.7 mg/mL for M_x mAb. Each arm consisted of 10 mice, five male, and five female. All male mice were sampled at 0.08 h, 72 h, and 336 h, while all female mice were sampled at 7 h, 168 h, and 672 h. Blood was collected under gas anesthesia by retroorbital sinus sampling in 1.5 mL microtubes without anticoagulant using microcapillaries. Sera were obtained by centrifugal spinning (13000 ×g for 2 min at room temperature) of the blood sample. Sera were stored at −80°C before dispatch and analysis.

For two animals from the CHO mAb KO group, one animal from the M_x mAb WT group, and for all five female animals from the M_x mAb KO group, tail vein injection was only partially successful and thus these animals showed a PK behavior mixed between IV and subcutaneous (SC) injection. For the glycoform-resolved PK parameter calculations, we excluded these animals. With respect to relative glycoform changes, we took all animals along, since we had shown previously that IV and SC injections behave consistently in this respect.[5] This was confirmed by the very high consistency of the relative abundances, additionally secured through an outlier analysis.

mAb and mIgG purification

mAbs were purified from mouse serum similarly to previous studies in other animal models.[4] In short, 5 µL of mouse plasma were diluted in 15 µL phosphate-buffered saline (PBS), captured for 1 h at room temperature with 2 µL CaptureSelect™ FcXL affinity matrix (Thermo Fisher Scientific), washed with PBS and water, and eluted with 100 mM formic acid.

For the assessment of mouse antibody glycosylation, total serum antibodies were purified using 15 µL of Protein G Sepharose beads (GE Healthcare). Though mAbs are co-captured, human and mouse glycopeptides are well separated chromatographically, and, hence, mAb glycopeptides do not interfere with liquid chromatography–mass spectrometry (LC-MS) quantitation of mouse glycopeptides. 2 µL of mouse serum was diluted in 100 µL PBS. All other steps of the capturing protocol were the same as for mAbs isolation.

LC-MS glycopeptide analysis

mAbs and mouse IgG Fc glycopeptides were analyzed with the GLycoLISA platform for which a detailed protocol is published.[16] Detailed approaches and analytes for the specific applications were described previously.[5,28]

Samples were buffer exchanged to 50 mM ammonium bicarbonate through drying, and incubated with 200 ng sequencing grade trypsin overnight. Resulting glycopeptides were analyzed by injecting 200 nL of the sample into a binary gradient at 700 nL/min supplied by an Ultimate 3000 RSLCnano LC system (ThermoFisher Scientific). Separation was performed on an RP-nanoLC Acclaim PepMap 100 C18 column 150 × 0.075 mm with 3 µm particles (ThermoFisher Scientific) with the previously published gradient.[5] Finally, analytes were detected on-line with a maXis™ quadrupole-time of time-of-flight mass spectrometer equipped with a nanoBooster™ nanoESI source (Bruker Daltonics, Bremen, Germany).

Processing of LC-MS data

Raw LC-MS data in mzXML format was processed using LaCyTools (version 2.0.1, build 20,201,216). Extraction and calibration of mAb glycopeptides in 2+ and 3+ charge states (retention time window 50–66 seconds) was performed using the parameters as specified in the GlYcoLISA protocol, with an integration mass window of 0.04 Th.

Curation of mAbs data

Analyte curation was first performed based on direct digests of the CHO mAb and M_x mAb, using the following quality requirements: absolute mass error <20 ppm, IPQ $\leq$ 0.2, S/N > 9. Analytes that fulfilled these quality criteria in the WT or M5 mAb in at least one charge state were included in the analysis with

both charge states. This consensus list of 20 analytes, was subsequently refined in lowest concentration spike samples of both mAbs (500 ng spike M5, 700 ng spike WT). Six analytes were retained for the M_x mAb, and nine analytes for the CHO mAb. In addition, for the M_x mAb, Man7-9 (2+ and 3+) could be quantified only in the first timepoint as a result of the expected circulating mannosidase action. Similarly, Man6 (2+) and Man8 (2+) were included for the first timepoint of the CHO mAb samples.

Curation of mIgG data

The mouse samples were separated into four biological groups, based on the mouse type (WT or KO) and the mAb type (CHO or M_x mAb). Analyte curation was performed per biological group and per charge state (2+ and 3+). An analyte was included in total area normalization when it fulfilled three quality criteria (same as for the mAbs) in 80% of the samples in at least one of the biological groups.

Antibody concentration measurements

mAb concentrations in the mouse serum samples were assessed with an ELISA method specific for the human IgG kappa chain.[4]

Non-compartmental pk analysis

Relative glycosylation profiles from LC-MS and absolute total mAb concentrations from ELISA were combined per animal and time point into individual glycoform concentrations as previously reported.[4] Total and glycoform-resolved PK parameters were obtained from these individual concentrations after dose adjustment (Figure S1, Table S12) by non-compartmental analysis, employing the Phoenix® WinNonlin program (NY, USA) for the kinetic evaluation. Visualization and statistical treatment of PK parameters and glycosylation profiles were done in GraphPad Prism 10.2.3 (GraphPad Software, San Diego, US).

Statistical analysis

Glycosylation traits of mIgG were compared between WT and KO mice, as well as between CHO and M_X mAb groups, using Welch's t-tests. Obtained *p*-values were corrected per subclass using the Benjamini–Hochberg method with a false discovery ratio (FDR) level of $\alpha = 0.05$.

PK parameters of the mAbs were compared within each group of animals using the same method. Comparison of the total mAb parameters between groups also employed Welch's t-test, but omitted FDR correction due to the low number of comparisons.

Abbreviations

CHO	Chinese hamster ovary
CQA	Critical quality attribute
CTLs	C-type lectins
Fc	fragment crystallizable
Gal	Galactose
GlcNAc,	*N*-Acetylglucosamine
LC	Liquid Chromatography
mAbs	(therapeutic) monoclonal antibody
MS	Mass spectrometry
MRC1	Mannose receptor C-type 1, also known as mannose receptor
MRC1 KO	Mannose receptor C-type 1 deficiency
TFA	Trifluoroacetic acid
WT	Wild type

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Author contributions

CRediT: **David Falck:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Steinar Gijze:** Data curation, Formal analysis, Validation, Visualization, Writing – review & editing; **Anna M. Wasynczuk:** Investigation, Validation, Writing – review & editing; **Sebastian Malik:** Investigation, Resources, Writing – review & editing; **Aline Tridon:** Investigation, Resources, Supervision, Writing – review & editing; **Martin Lechmann:** Conceptualization, Data curation, Formal analysis, Validation, Writing – review & editing; **Manfred Wuhrer:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing; **Dietmar Reusch:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Disclosure statement

Sebastian Malik, Martin Lechmann, and Dietmar Reusch are employees of Roche Diagnostics GmbH. The other authors have no conflicting interests to declare.

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Data availability statement

Data underlying all visualizations can be found in the Supplemental Data.

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